

AI Agent Bootcamp - Problem Statement & Project Briefs

1. 🏠 Smart Hostel Management Agent

1.1 🎯 Theme

Smart Hostel Management Agent

1.2 📄 Project Title

HostelCare AI — Intelligent Hostel Operations & Complaint Management Agent

1.3 🔍 Problem Breakdown

Students frequently face issues related to **room maintenance, electricity, water supply, Wi-Fi, plumbing, cleanliness, furniture, and other hostel facilities.**

Hostel complaints are often handled through registers, phone calls, WhatsApp messages, or informal communication. This can result in:

- Complaints being missed or forgotten
- Delayed maintenance
- Incorrect assignment of complaints
- No clear priority for urgent issues
- Students having no visibility into complaint status
- Repeated complaints about the same underlying problem

For example:

A student reports that the bathroom tap in Room 204 has been leaking for three days.

Instead of simply recording the complaint, an intelligent system should understand the issue, determine its category and urgency, and initiate the appropriate resolution process.

1.4 🤖 AI Agent's Role

The AI Agent should act as an **intelligent hostel operations coordinator.**

It should:

- Understand complaints submitted in natural language.
- Extract relevant information such as hostel block, room number, and issue type.
- Categorize and prioritize complaints.
- Assign complaints to the appropriate maintenance team.

- Generate and track maintenance tickets.
- Notify students about status changes.
- Identify recurring problems across hostels.

1.5 Key Capabilities & Features

1.5.1 Intelligent Complaint Registration

- Accept text and optionally image-based complaints.
- Extract room/block information.
- Categorize complaints as plumbing, electrical, Wi-Fi, cleaning, etc.

1.5.2 Priority & Assignment

- Determine low, medium, or high priority.
- Identify the appropriate maintenance department.
- Assign the complaint automatically.

1.5.3 Complaint Tracking & Escalation

- Generate a unique ticket ID.
- Allow students to check complaint status.
- Escalate unresolved complaints after a defined period.

1.5.4 Hostel Analytics

- Detect recurring complaints.
- Identify problematic hostel blocks or facilities.
- Generate maintenance reports for administrators.

1.6 Potential Data Sources & Tools

- Hostel/student database
- Maintenance staff database
- Complaint/ticket database
- Hostel policies
- Google Sheets
- MongoDB/PostgreSQL
- REST APIs

- Notification/email APIs
- Google ADK, LangGraph, LangChain, CrewAI, or similar frameworks
- Vision models for image-based complaints

1.7 🚀 Expected Output / Success Criteria

A working MVP should demonstrate:

Student complaint → Agent understands issue → Classifies → Prioritizes → Creates ticket → Assigns team → Tracks status → Provides resolution update

A simulated maintenance backend is acceptable for the prototype.

1.8 ⚠️ Potential Challenges

- Correctly determining complaint priority.
- Maintaining reliable ticket state and preventing duplicate complaints.

2. 🚌 Smart Student Transport Agent

2.1 🎯 Theme

Smart Student Transport Agent

2.2 📄 Project Title

CampusMove AI — Intelligent Student Transport & Route Agent

2.3 🔍 Problem Breakdown

Students frequently struggle with:

- Bus timings
- Routes and stops
- Delays
- Crowded buses
- Route changes
- Missed buses
- Finding the fastest or most convenient route

Transport information may be distributed across timetables, notices, spreadsheets, or messaging groups.

For example:

"I need to reach college by 9:00 AM from the hostel."

The student needs more than a timetable. They need the system to **analyze available options and recommend the best one.**

2.4 🤖 AI Agent's Role

The agent should act as an **intelligent campus mobility planner.**

It should understand:

- Starting location
- Destination
- Required arrival time
- Available transportation
- Delays or route changes
- Travel duration

It should then compare available options and recommend the most suitable one.

2.5 🛠️ Key Capabilities & Features

2.5.1 Route & Schedule Retrieval

- Search available buses.
- Retrieve routes and stops.
- Check schedules.

2.5.2 Intelligent Route Recommendation

Compare options based on:

- Arrival time
- Travel duration
- Cost
- Convenience
- Availability

2.5.3 Delay & Alternative Handling

If the preferred bus is unavailable or delayed:

Detect change → Find alternatives → Recommend another route.

2.5.4 Personalized Trip Planning

The agent can consider the student's required arrival time and recommend an appropriate departure time with a safety buffer.

2.6 📁 Potential Data Sources & Tools

- Campus bus schedules
- Route/stop database
- GPS/location APIs
- Maps APIs
- OpenStreetMap
- Traffic APIs
- Weather APIs
- Google Sheets/database
- Notification APIs

2.7 🚀 Expected Output / Success Criteria

A working MVP should demonstrate:

Origin + Destination + Required arrival time → Retrieve transport options → Compare → Recommend best option → Explain recommendation

A strong demo can simulate a bus delay and show the agent automatically selecting an alternative.

2.8 ⚠️ Potential Challenges

- Real-time transport and traffic data may not always be available.
- Outdated schedules can produce incorrect recommendations.

3. 📢 Civic Complaint & Resolution Agent

3.1 🎯 Theme

Civic Complaint & Resolution Agent

3.2 📄 Project Title

CivicResolve AI — Intelligent Public Issue Reporting & Resolution Agent

3.3 🔍 Problem Breakdown

Citizens encounter problems such as:

- Potholes
- Garbage accumulation
- Broken streetlights
- Drainage issues
- Water supply problems
- Damaged public infrastructure

Citizens often don't know:

- Which authority handles the issue
- What category the complaint belongs to
- How urgent it is
- What information needs to be submitted
- How to track the complaint

Authorities also receive poorly structured complaints, making prioritization and routing difficult.

3.4 🤖 AI Agent's Role

The agent should act as an **intelligent civic issue coordinator**.

It should:

- Understand complaints submitted through text or images.
- Identify the type of civic problem.
- Extract or validate location information.
- Determine priority.
- Identify the responsible department.
- Generate a structured complaint.
- Track its resolution.
- Escalate unresolved complaints.

3.5 🛠️ Key Capabilities & Features

3.5.1 Multimodal Complaint Understanding

- Accept text, images, and location.
- Analyze uploaded images where applicable.

3.5.2 Intelligent Classification

Classify issues such as:

- Roads
- Garbage
- Drainage
- Water
- Streetlights
- Public infrastructure

3.5.3 Authority Routing

Determine which department or authority should receive the complaint.

3.5.4 Resolution Tracking

- Generate complaint ID.
- Track status.
- Send reminders.
- Escalate unresolved issues.

3.6 Potential Data Sources & Tools

- Civic complaint database
- Municipal department information
- GIS/location data
- Maps APIs
- Vision models
- Government open-data portals
- Database
- Email/SMS/notification APIs

3.7 🚀 Expected Output / Success Criteria

A working MVP should demonstrate:

Image/Text complaint → Identify issue → Extract location → Categorize → Determine priority → Route to authority → Generate tracking ID

A simulated municipal backend can be used for demonstration.

3.8 ⚠️ Potential Challenges

- Images may not provide enough information for accurate classification.
 - Incomplete location information can make authority routing difficult.
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4. 🖥️ Campus IT Support Agent

4.1 🎯 Theme

Campus IT Support Agent

4.2 📄 Project Title

CampusFix AI — Autonomous IT Troubleshooting & Support Agent

4.3 🔍 Problem Breakdown

Students and faculty frequently experience:

- Wi-Fi problems
- Login issues
- Password problems
- Software installation issues
- Network connectivity problems
- Printer problems
- System configuration issues

IT teams often spend significant time resolving repetitive first-level problems.

This leads to:

- Increased IT workload
- Longer response times
- Repeated support requests

- Poor issue categorization
- Difficulty tracking unresolved issues

4.4 🤖 AI Agent's Role

The agent should function as a **first-level autonomous IT support assistant**.

Rather than immediately creating a ticket, it should:

1. Understand the problem.
2. Ask diagnostic questions.
3. Search the IT knowledge base.
4. Recommend troubleshooting steps.
5. Determine whether the problem is resolved.
6. Create a ticket if required.
7. Escalate complex issues.

4.5 🛠️ Key Capabilities & Features

4.5.1 Conversational Diagnosis

The agent should ask follow-up questions based on the user's answers.

Example:

Student: "Campus Wi-Fi isn't working."

Agent:

"Are other devices able to connect to the same network?"

4.5.2 Knowledge Retrieval

Retrieve relevant information from:

- IT FAQs
- Troubleshooting manuals
- University policies
- Internal documentation

4.5.3 Automated Troubleshooting

Provide safe, step-by-step solutions for common problems.

4.5.4 Ticket Creation & Escalation

If the problem cannot be resolved:

Diagnosis → Ticket creation → Assignment → Tracking → Escalation

4.6 Potential Data Sources & Tools

- University IT knowledge base
- FAQ documents
- Troubleshooting manuals
- Ticket database
- Network-status APIs
- RAG/vector database
- Google ADK
- LangGraph
- LangChain
- CrewAI

4.7 Expected Output / Success Criteria

MVP:

Student reports problem → Agent diagnoses → Retrieves solution → Guides troubleshooting → Checks result → Creates ticket if unresolved

This project provides a strong opportunity to demonstrate **RAG + tool usage + agent reasoning**.

4.8 Potential Challenges

- Preventing the agent from recommending unsafe system/network changes.
- Ensuring retrieved troubleshooting information is accurate and relevant.

5. Smart Waste Management Agent

5.1 Theme

Smart Waste Management Agent

5.2 Project Title

EcoSort AI — Intelligent Waste Classification & Management Agent

5.3 Problem Breakdown

People often don't know how to correctly:

- Identify waste categories
- Segregate waste
- Dispose of electronic waste
- Recycle materials
- Locate collection facilities

Incorrect waste disposal contributes to:

- Environmental pollution
- Poor recycling efficiency
- Increased waste-management costs
- Unsafe handling of certain waste

This is especially relevant to **universities, hostels, residential communities, and public spaces.**

5.4 AI Agent's Role

The agent should act as an **intelligent waste-management advisor.**

It should determine:

- What type of waste is present
- How it should be segregated
- How it should be disposed of
- Where it can be taken
- Whether special handling is required

5.5 Key Capabilities & Features

5.5.1 Waste Identification

Identify categories such as:

- Organic
- Plastic
- Paper
- Glass

- Metal
- E-waste

5.5.2 Disposal Recommendation

Recommend the appropriate disposal or recycling method.

5.5.3 Facility Locator

Identify nearby or campus-based collection/recycling facilities.

5.5.4 Waste Analytics

For institutions, analyze:

- Frequently generated waste
- High-waste locations
- Collection requirements
- Potential improvement areas

5.6 Potential Data Sources & Tools

- Waste classification datasets
- Vision models
- Campus waste database
- Recycling facility database
- Maps/location APIs
- Local waste-management rules
- Environmental open-data sources

5.7 Expected Output / Success Criteria

MVP:

Upload waste image → Identify category → Explain segregation → Recommend disposal method → Locate collection facility

A second workflow can demonstrate waste analytics for a campus administrator.

5.8 Potential Challenges

- Images may contain multiple types of waste.
- Disposal rules vary by location and must be based on reliable local information.

6. 🧑🌾 Smart Farmer Assistance Agent

6.1 🎯 Theme

Smart Farmer Assistance Agent

6.2 📄 Project Title

AgriGuide AI — Intelligent Farmer Assistance & Advisory Agent

6.3 🔍 Problem Breakdown

Farmers often need timely information about:

- Weather
- Crop practices
- Government schemes
- Agricultural resources
- Market information
- Pest/disease concerns
- Irrigation
- Crop planning

Useful information may be distributed across different sources and can be difficult to interpret.

For example:

"Is this week suitable for working on my crop?"

Answering such a question may require combining **location + crop information + weather data + reliable agricultural guidance**.

6.4 🤖 AI Agent's Role

The agent should function as an **agricultural information and decision-support assistant**.

It should:

1. Understand the farmer's question.
2. Identify crop, location, and context.
3. Retrieve relevant information.
4. Compare information from trusted sources.

5. Provide an understandable recommendation.
6. Identify relevant government schemes.
7. Explain the sources behind its response.

6.5 Key Capabilities & Features

6.5.1 Weather-Aware Assistance

Retrieve relevant:

- Current weather
- Forecast
- Rainfall
- Temperature
- Other weather indicators

6.5.2 Crop Assistance

Provide information regarding:

- General cultivation practices
- Irrigation
- Crop management
- Common agricultural concerns

6.5.3 Government Scheme Discovery

The agent can:

Farmer profile → Search schemes → Check basic eligibility → Identify documents → Explain application process

6.5.4 **Market & Resource Information**

Retrieve relevant market or agricultural-resource information and present it in simple language.

6.6 Potential Data Sources & Tools

- Government agriculture portals
- Trusted agricultural information sources
- Weather APIs
- Agricultural datasets
- Market-price APIs where available
- Location/GIS APIs

- RAG/document retrieval
- LLM
- Google ADK
- LangGraph
- LangChain
- CrewAI

6.7 🚀 Expected Output / Success Criteria

MVP:

Farmer provides crop + location + question → Agent retrieves relevant information → Reasons over the information → Provides understandable guidance → Identifies information sources

A second workflow can demonstrate:

Farmer profile → Relevant government schemes → Basic eligibility → Required documents → Application guidance

6.8 ⚠️ Potential Challenges

- Agricultural recommendations can be highly crop- and region-specific.
- Outdated or unreliable information can lead to poor decisions, so source verification and uncertainty handling are essential.